

Generic Drug Prices and the Number of Firms: Preliminary Descriptive Analysis

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Abstract

We document how generic drug prices in the United States co-move with the number of generic suppliers, using IQVIA quarterly data for 2004–2019. Three findings. First, the trajectory of the generic firm count within a molecule commonly follows a rise–peak–decline pattern: among molecules whose count peaks at three or more firms, roughly one third lose at least half of their suppliers by the end of the sample, and only one in ten remains at its peak. Second, generic prices move asymmetrically over this cycle: molecules with substantial firm-count declines see average generic prices *rise* by about 0.9–1.2 log points from the peak year onward, whereas prices are flat where counts hold. Third, the price reversal after exit is incomplete: comparing periods with the same number of firms before and after the peak, and controlling for aggregate trends, roughly half of the competitive price reduction achieved at the peak persists after the firm count falls back (“price hysteresis”), with a conservative lower bound of about one quarter.

1 Data and panel construction

Source data. Four IQVIA extracts of US sales at the country–sector–corporation–product–pack–quarter level: 2004–2008, 2009–2010, 2011–2016 (Excel), and 2017–2019 (csv). Each records manufacturer revenue (LCD MNF, equal to USD for the US) and standard units. Standard units (*SU*) are pack-size adjusted: one *SU* equals one tablet/capsule, verified in the raw data ($SU = \text{packs} \times \text{pack size}$ holds row by row where both are observed, and implied unit prices are plausible, e.g. Lipitor 20mg at \$2.79 per tablet in 2004Q1). Summing *SU* across pack sizes is therefore valid.

Cleaning. We keep single-ingredient products (fixed-dose combinations are dropped; in the raw files FDC rows are duplicated once per ingredient—and per salt in 2011–2016—so we also de-duplicate identical rows to avoid double counting). Each extract is collapsed to cells defined by corporation $\times$ molecule $\times$ strength $\times$ quarter, separately for branded and generic products. A product is classified generic if (a) its name contains the molecule name, (b) its first word is a truncation of the molecule name (IQVIA cuts long names to about 13 characters, e.g. METO-CLOPRAMID TEVA), (c) it carries IQVIA’s unbranded-commodity suffix (STCE), or (d) its first word is shared by three or more corporations within the molecule, which catches drug abbreviations (HCTZ) and store-brand OTC names, or (e) its name contains the corporation’s name or 4-character code (MKES = McKesson) and its first word is shared by two or more corporations, which catches synonym-named generics (ALBUTEROL under molecule SALBUTAMOL, ASPIRIN under ACETYLSALICYLIC ACID) while leaving single-source brands that carry a code (ZYRTEC J.J.) branded; branded otherwise (e.g. LIPITOR, REGLAN). The four cleaned segments are appended

into a panel of 1,110,905 cells. Branded products account for 31% of cells but 87% of revenue (\$4.23T of \$4.85T), consistent with the classification behaving sensibly.

Molecule panel. The analysis panel collapses to molecule $\times$ quarter, excluding IQVIA’s catch-all pseudo-molecules (diagnostic kits, medical devices, cream bases, “composition unknown”): 97,608 observations on 2,210 molecules, 2004Q1–2019Q4 ($\approx$ 1,450–1,650 molecules active per quarter). Table 1 reports summary statistics. Firm counts are counts of distinct corporations with positive sales; prices are revenue per standard unit.

Table 1: Molecule–quarter panel: summary statistics

	Obs.	Mean	Std. dev.	Min	Max
No. generic firms	97,608	3.33	5.39	0	43
No. branded firms	97,608	1.56	2.13	0	34
Generic price (\$/ <i>SU</i>)	51,946	23.4	165.2	0.001	7,579
Branded price (\$/ <i>SU</i>)	83,395	394.9	9,820	0.001	2,054,926

Notes: One observation is a molecule–quarter. Prices are total revenue divided by total standard units, defined when the corresponding volume is positive. Molecules with many branded firms are over-the-counter products with genuinely many brand names (e.g. diphenhydramine: Benadryl, Unisom, Nytol); prescription molecules almost always have at most one or two.

2 Linking the four extracts

Firm names are stamped with the corporate structure as of each extract’s pull date, and pull dates are not chronological across extracts (e.g. WYETH and SCHERING PLOUGH disappear at the 2008/09 seam while ABBVIE and BAUSCH HEALTH—names that did not exist until 2013 and 2018—appear; BAUSCH HEALTH then reverts to VALEANT at the 2010/11 seam; VIATRIS and ORGANON appear at the 2016/17 seam). Firm-level linking across extracts is therefore unreliable, while molecule-level aggregates are clean. Table 2 quantifies this: apparent exit rates of firm–molecule–strength cells jump an order of magnitude at the three extract boundaries relative to identical seasonal transitions within extracts, with mirror-image entry spikes. At the molecule level, by contrast, coverage moves smoothly, within-molecule changes in generic firm counts show only small boundary effects (mean Δn of -0.26 to $+0.20$ at boundaries vs. about $+0.05$ at normal Q4→Q1 transitions), and generic price changes at boundaries are indistinguishable from normal quarters. The analysis is accordingly conducted at the molecule level, and event-style analyses treat the three boundary transitions as non-informative.

Validation. The panel reproduces the canonical loss-of-exclusivity pattern. For atorvastatin (Lipitor, LOE November 2011): three generic corporations enter in 2011Q4 at \$3.21 per *SU* against the brand’s \$4.16; by 2012Q4 twelve generic firms sell at \$0.28; by 2019 about 21 firms sell at \$0.07–0.09, while the branded price *rises* from \$4 to \$9.61.

3 Trajectories of the generic firm count

Is rise–peak–substantial-decline a common firm-count trajectory? For each molecule we compute annual firm counts (the maximum across the year’s quarters, which absorbs boundary noise), the

Table 2: Apparent exit and entry rates at extract boundaries

Transition (Q4→Q1)	Exit rate	Entry rate
2008Q4 → 2009Q1 (boundary)	0.444	0.407
2010Q4 → 2011Q1 (boundary)	0.237	0.264
2016Q4 → 2017Q1 (boundary)	0.365	0.337
Mean, 12 within-extract Q4→Q1	0.028	0.039
Mean, all non-boundary quarters	0.030	–

Notes: Cells are corporation–molecule–strength combinations with positive standard units. Exit: active in t , not active in $t+1$. Entry: active in t , not in $t-1$ (dated by the quarter entered). Boundary spikes reflect IQVIA’s retroactive renaming of corporations, not real firm turnover.

peak count, and the end-of-sample count. Table 3 classifies molecules. Among the 497 “judgeable” molecules—peak of at least 3 generic firms, reached at least three years before the end of the sample, observed through 2019—the median molecule ends at 67% of its peak count; one third lose at least half their suppliers; only 10% remain at peak. The peak follows a genuine rise (median +2 firms relative to the first observed year, mean +4.3; 73% of peaks are interior). Declines are strongly concentrated in smaller markets: 54% of molecules peaking at 3–5 firms fall substantially, versus 16% of those peaking at 11+. Excluding vitamins/minerals (ATC A11/A12) changes nothing.

Table 3: Trajectory classification of generic firm counts, by molecule

	Molecules	Share
No generics ever	975	44.1%
Low activity (peak 1–2 firms)	381	17.2%
Leaves sample before 2019	60	2.7%
Peak <3 years before end (censored)	297	13.4%
<i>Judgeable (peak ≥ 3, $\geq 3y$ post-peak):</i>	497	
Substantial fall (end $\leq 50\%$ of peak)	170	34.2%
Moderate fall (50–75%)	139	28.0%
Small fall (75–99%)	137	27.6%
No fall (ends at peak)	51	10.3%

Notes: Annual firm count = maximum over the year’s quarters. Shares in the top panel are of all 2,210 molecules; shares in the bottom panel are of the 497 judgeable molecules. Molecules that leave the sample entirely (the ultimate fall) are kept as a separate category, so the fall rates below are conservative.

Two checks confirm the declines are real rather than artifacts of the extract seams. First, within the single-vintage 2011–2016 extract (molecules peaking 2011–13, observed through 2016), the median molecule ends at 80% of peak over just a three-year window, with 12% at $\leq 50\%$. Second, the event-time profile around interior peaks (Figure 1) shows a symmetric climb from 43% of peak six years before to the peak, followed by a decline to 69% six years after; the first post-peak year is partly mechanical (the peak is a maximum), but the continued monotone decline through year +6 is not.

A cleaner cohort view aligns molecules at their *first* generic entry (260 episodes in 2006–2015 with a branded incumbent and at least one generic-free year of prior history; no mechanical peak alignment). The mean firm count rises from 2.3 at entry to 6.1 after four years and is still rising at

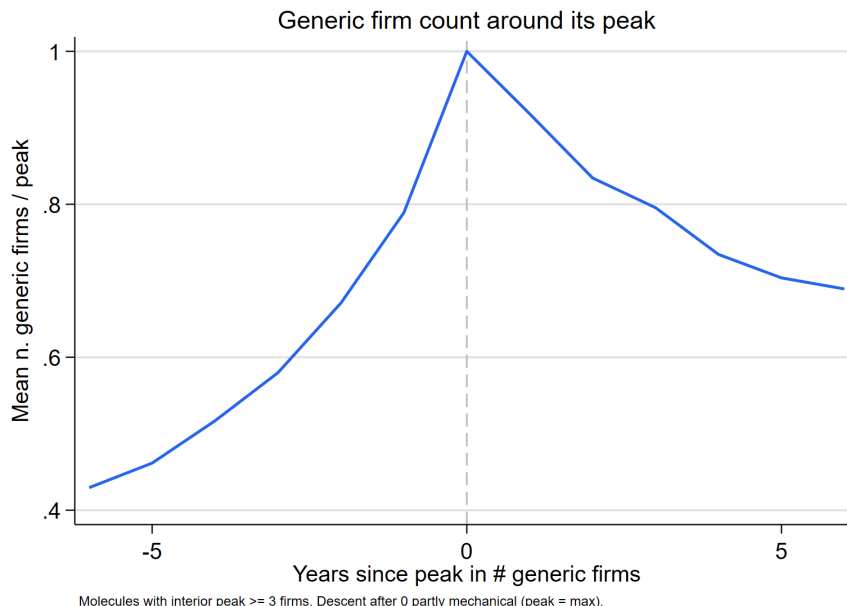


Figure 1: Generic firm count around its peak (interior peaks ≥ 3 firms; count normalized by the molecule’s peak).

that horizon, while the median generic price falls to 48% of its at-entry level (Figure 2). Declines thus typically arrive later in the market’s life cycle, though 15% of these young cohorts have already lost half of their peak count within four years.

4 Price patterns around firm-count declines

4.1 Prices rise where counts collapse

Table 4 relates the change in the average generic price from the peak year to the end of the sample to the trajectory class. Molecules with substantial firm-count declines see generic prices rise by a median 0.87 log points (mean 1.24); where counts hold, prices are flat to declining. The gradient is monotone.

Table 4: Change in log generic price, peak year $\rightarrow$ last year, by class

Trajectory class	N	Mean	Median	p75
Substantial fall	149	1.24	0.87	1.87
Moderate fall	139	0.27	−0.06	0.75
Small fall	137	0.10	−0.16	0.25
No fall	51	0.13	−0.08	0.23

Notes: Judgeable molecules with defined prices in both the peak and final year. Prices are average revenue per standard unit across the molecule’s generic products.

Several of the largest substantial-fall molecules by revenue are known regulatory episodes—colchicine (13 $\rightarrow$ 6 firms, price +4.5 log points), vasopressin (5 $\rightarrow$ 0), pancrelipase (4 $\rightarrow$ 0), all affected by the FDA Unapproved Drugs Initiative—while others (atenolol 20 $\rightarrow$ 10, bicalutamide 12 $\rightarrow$ 6,

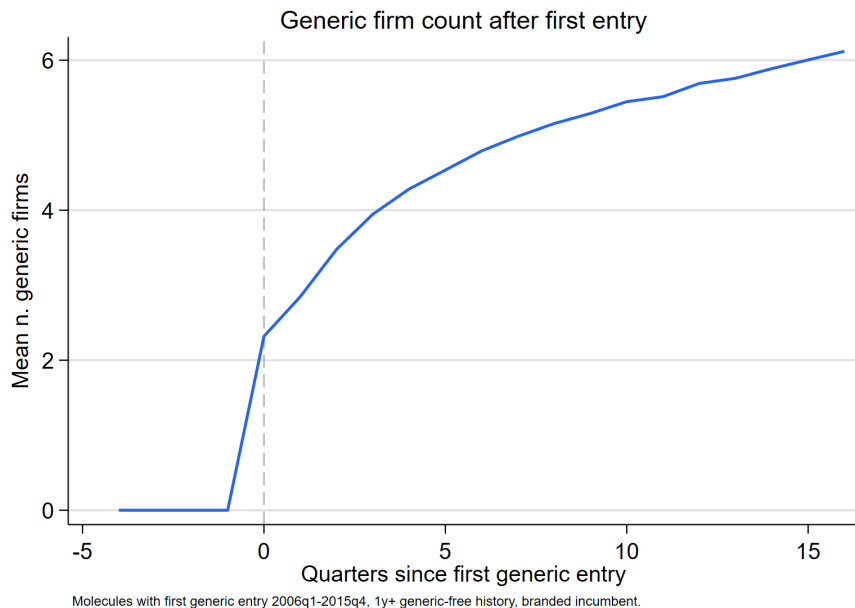


Figure 2: Generic firm count after the first generic entry (mean across 260 molecules; entries 2006Q1–2015Q4).

nicardipine 9→4) resemble ordinary competitive shakeouts. Regulatory removals are thus a quantitatively important and plausibly exogenous source of exit.

4.2 The reversal is incomplete: price hysteresis

If a molecule’s firm count rises from 5 to 10 and later falls back to 5, does the price return to its original 5-firm level? We estimate

$$\ln p_{mt} = \alpha_m + \delta_t + \beta \ln n_{mt} + \gamma \ln n_{mt}^{\max} + \varepsilon_{mt},$$

where n_{mt} is the current generic firm count and $n_{mt}^{\max}$ its within-molecule running maximum. Complete reversal implies $\gamma = 0$; the share of the up-leg price reduction retained after the count returns is $\gamma/(\beta + \gamma)$. Table 5 shows both current and historical-peak counts matter. The baseline retained share is 0.54 (s.e. 0.10); allowing molecule-specific linear trends—a conservative specification, since trends absorb genuine long-run hysteresis—gives an imprecise 0.27 (s.e. 0.16); the peak- ≥ 5 sample gives 0.65. In levels, the current count is largely priced out (-0.021 , marginal) while the historical peak carries -0.117 per firm: the competitive effect of the *current* count is concentrated at low n , and history dominates elsewhere.

At the baseline estimates, the worked example reads: moving from 5 to 10 firms lowers price by 0.51 log points; after the count returns to 5, price sits 0.28 log points *below* its original 5-firm level—about 54% of the reduction is retained.

A model-free check compares, within molecule, the quarter-FE-adjusted log price in quarters with exactly n firms before versus after the peak (1,143 molecule $\times$ n pairs, peaks ≥ 4): the post-peak price is lower in 89% of pairs, by a median 0.63 log points, and the gap widens the further the count sits below its peak (medians -0.50 , -0.95 , and -1.22 log points at 1–2, 3–5, and 6+ firms below peak). Taken at face value this implies more-than-complete retention; because this

Table 5: Price hysteresis: current vs. historical-peak firm count

	(1) Baseline	(2) Mol. trends	(3) Peak ≥ 5	(4) Levels
$\ln n$	−0.341 (0.066)	−0.303 (0.052)	−0.304 (0.092)	
$\ln n^{\max}$	−0.401 (0.088)	−0.114 (0.076)	−0.575 (0.119)	
n				−0.021 (0.013)
$n^{\max}$				−0.117 (0.015)
Retained share $\gamma/(\beta+\gamma)$	0.54 (0.10)	0.27 (0.16)	0.65 (0.11)	0.85 (0.09)
Molecule FE, quarter FE	yes	yes	yes	yes
Molecule trends	no	yes	no	no
Observations	51,926	51,926	34,406	51,926
Molecules	1,215	1,215	666	1,215

Notes: Dependent variable: log average generic price per standard unit. $n^{\max}$ is the within-molecule running maximum of the generic firm count. Standard errors clustered by molecule. Sample: molecule–quarters with at least one generic firm and positive generic volume.

comparison absorbs no molecule-specific life-cycle price trend, we read it as an upper bound and the trend-adjusted regression as the preferred estimate.

Two forces bias *against* finding retention, making these estimates conservative: if low-price suppliers are the ones that exit, the surviving-firm average mechanically rises on the down leg; and generic prices trend downward over the molecule life cycle only insofar as that trend is absorbed by specification (2).

5 Caveats and next steps

- **Composition.** Prices are molecule-level averages across strengths and products; hysteresis should be re-estimated within strength/pack using the corporation-level files (feasible within extract segments) to separate surviving-firm price stickiness from composition.
- **Endogeneity.** Entries and exits are equilibrium outcomes. Regulatory removals (Unapproved Drugs Initiative, import bans) provide plausibly exogenous exit variation and already appear among the largest declines in the data.
- **Collusion.** The 2013–14 generic price-fixing episode raised prices without exit and should be handled explicitly in any causal design.
- **Measurement.** IQVIA manufacturer revenue is invoice-based (rebates are modest for generics); firm counts inherit IQVIA’s corporate-name vintages, so firm-level dynamics are only used within extract segments; branded firm counts are inflated for OTC molecules by store-brand products.